

2 Description

The STM32C051x6/x8 mainstream microcontrollers are based on high-performance Arm® Cortex®-M0+ 32-bit RISC core operating at up to 48 MHz frequency. Offering a high level of integration, they are suitable for a wide range of applications in consumer, industrial and appliance domains and ready for the Internet of Things (IoT) solutions.

The devices incorporate a memory protection unit (MPU), high-speed embedded memories (12 Kbytes of SRAM and up to 64 Kbytes of flash program memory with read and write protection, and securable area), DMA, an extensive range of system functions, enhanced I/Os, and peripherals. The devices offer standard communication interfaces (two I²Cs, two SPI / one I²S, and two USARTs), one 12-bit ADC (2.5 MSps) with up to 21 channels, a low-power RTC, an advanced control PWM timer, four general-purpose 16-bit timers, a 32-bit general-purpose timer, two watchdog timers, and a SysTick timer.

The devices operate within ambient temperatures from -40 to 125°C and with supply voltages from 2.0 V to 3.6 V. Optimized dynamic consumption combined with power-saving modes allows the design of low-power applications.

The devices are housed in packages with 15 to 48 pins.

Table 2. STM32C051x6/x8 family device features and peripheral counts

Peripheral		STM32C051_								
		_D8	_F6	_F8	_G6	_G8	_K6	_K8	_C6	_C8
Flash memory (Kbyte)		64	32	64	32	64	32	64	32	64
SRAM (Kbyte)		12 with parity								
Timers	Advanced control	1 (16-bit)								
	General-purpose	1 (32-bit)								
		4 (16-bit)								
	SysTick	1								
	Watchdog	2								
Comm. interface	SPI [I2S] ⁽¹⁾	2 [1] + 2 extra through USARTs								
	I2C	2								
	USART	2								
RTC		Yes								
GPIOs (all 5V-tolerant)		13	18		26		30		45	
DMA channels		5								
Wakeup pins		4	4		5		5		5	
12-bit ADC channels (ext. + int.)		9 + 4	13 + 4		15 + 4		16 + 4		19 + 4	
Max. CPU frequency		48 MHz								
Operating voltage		2.0 to 3.6 V								
Operating temperature ⁽²⁾		Ambient: -40 to 85 °C / -40 to 105 °C / -40 to 125 °C Junction: -40 to 105 °C / -40 to 125 °C / -40 to 130 °C								
Packages		WLCSP15	TSSOP20		UFQFPN28		LQFP32 UFQFPN32		LQFP48 UFQFPN48	
Bootloader		USART1, USART2, I2C1, I2C2, SPI1, SPI2								

1. The numbers in brackets denote the count of SPI interfaces configurable as I²S interface.

2. Depends on order code. Refer to [Section 7: Ordering information](#) for details.